

Sunny Bagal

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Profile

2026 Computer Engineering graduate with internship experience leading backend development for a four-person team. Built and deployed a RAG application and a real-time collaborative whiteboard, with correctness verified through testing and performance improvements measured through reproducible benchmarks.

Skills

Languages: Python, TypeScript, JavaScript, SQL

Backend: Node.js, Bun, Express, REST APIs, WebSockets, SSE, BullMQ, JWT, OAuth 2.0, Prisma, Drizzle

Databases: PostgreSQL (SQL, pgvector), Redis (caching), MongoDB (NoSQL)

Cloud & DevOps: Railway, Render, AWS (EC2, S3), DigitalOcean (Droplets), Docker, Linux, Git, GitHub Actions

Testing: Vitest, Mocha, Supertest, React Testing Library, integration and load testing

AI/ML: RAG, embeddings, vector search (pgvector, HNSW), hybrid retrieval, LLM APIs (Claude, OpenAI)

Frontend: React, Next.js, Tailwind CSS

CS Fundamentals: DSA, OOP, DBMS, operating systems, computer networks, complexity analysis

Experience

IntelCorp Software India Pvt. Ltd.

Jan 2026 – Apr 2026

Software Engineering Intern

Navi Mumbai

- Led technical work for a four-person intern team building GeoGrid. Owned the Node.js/Express backend and integrated frontend, enrichment, and billing modules behind a unified API.
- Designed recursive geospatial grid subdivision with polygon-aware boundary filtering. Cells reaching the provider's per-query result limit split into four sub-cells, recovering POIs hidden by single-area searches.
- Designed and benchmarked a Redis cell cache on a seeded 5,000-POI test set: repeat searches made zero provider API calls (down from 296), and 50%-overlap searches made 51.7% fewer.

Projects

Linea – Real-time collaborative whiteboard

Jul 2026

TypeScript, Next.js, Turborepo, PostgreSQL, Prisma, Canvas 2D, Railway

[Code](#) | [Live](#)

- Built the WebSocket sync layer from scratch (no Socket.IO): JWT authentication, room management, exponential-backoff reconnection, and a typed, Zod-validated message protocol.
- Designed a pure fold over an append-only, server-sequenced operation log. One reducer drives live sync, reload hydration, and read-only time-travel; undo appends compensating ops without rewriting history.
- Found that board loading scaled quadratically with history size. A linear-time reducer cut a 10,000-operation replay from 185 ms to 0.53 ms, and 100,000-operation boards now load in under 8 ms.
- Correctness-gated load tests sustained 400 ops/sec across 8 clients at ~5–6 ms p95 with zero loss, duplication, or reordering. Traced the 800+ ops/sec ceiling to queueing on the per-op sequencing transaction.

Recall – AI-powered link organizer

May 2026

TypeScript, Node.js, PostgreSQL/pgvector, Drizzle, Redis, BullMQ, Claude API, React, Railway

[Code](#) | [Live](#)

- Built BullMQ workers to fetch, summarize (Claude API), and embed saved links into pgvector asynchronously, keeping ingestion work off the request path.
- Implemented hybrid retrieval without LangChain: pgvector HNSW + weighted tsvector/GIN FTS, fused via reciprocal rank fusion. Replacing ILIKE with GIN-indexed full-text search cut median latency from 28.4 to 7.9 ms.
- Built a retrieval benchmark across 1k–50k rows measuring recall@10 and query plans. A query-planner fix made vector search 20× faster at 10k rows (74.9 ms → 3.7 ms).
- Shipped SSE-streamed RAG chat grounded in the user's saved links, with citation rendering.

NeuralCompress – Dual-Mode Image Compression System

Jan 2025 – Jun 2025

PyTorch, CompressAI, Flask, Hugging Face Spaces

[Code](#) | [Live](#)

- Built a near-lossless image compression pipeline in PyTorch and shipped it as a Flask demo with a custom binary file format.
- Benchmarked a learned codec (CompressAI) against JPEG and WebP at matched quality: 2.2× smaller files (0.45 vs 0.98 bits per pixel).

Education

B.Tech. in Computer Engineering

2022 – 2026

Narsee Monjee Institute of Management Studies

Navi Mumbai

CGPA: 2.95/4